# Supplement to When Does Verification Help? Recovery-Path Regularization for Strategy-Aware Valuation

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This supplement reports additional fixed-design diagnostics and the matched stronger-gap comparison. It preserves the distinction between new training, new diagnostics from existing models, and the earlier four-stage audit. It does not establish new-market effectiveness or repair the retained strict cross-machine numerical failure. All models and per-listing artifacts remain local; released tables contain source-, seed- or batch-level aggregates.

## S1 Data construction and operational assumptions

**Sample flow.** Historical construction receipts bind the input versions and every split checksum. Joint filtering precedes non-price fingerprint deduplication; individual exclusion flags overlap and must not be added. The counts below reconcile all retained records, including calibration and test partitions. Only train and validation enter the reported fixed-cohort experiments; calibration and test are not newly evaluated here. Sealed test labels were hashed as bytes for integrity, not parsed for this revision. JUCars retains used-condition records with positive finite JOD price and

Table S1: Construction counts. Filtered counts are joint exclusions; deduplication follows filtering.

| Source | Raw | Filtered | Duplicates | Retained | Used train/eval |
|---|---|---|---|---|---|
| Synthetic | 7500 | 0 | 0 | 7500 | 5000/1000 |
| JUCars | 10921 | 2117 | 46 | 8758 | 4774/1000 |
| MUCars | 101896 | 28892 | 1693 | 71311 | 5000/1000 |

Table S2: Partitions before label-independent experimental truncation. Synthetic validation denotes its separately generated evaluation population.

| Source | Train | Validation | Calibration | Test |
|---|---|---|---|---|
| Synthetic | 6000 | 1500 | 0 | 0 |
| JUCars | 4774 | 1374 | 1343 | 1267 |
| MUCars | 39244 | 10672 | 10696 | 10699 |

model year 1980–2026. MUCars excludes records labeled New, restricts advertised MAD price to [5,000,2,000,000], and model year to 1980–2025. These are inherited quality filters, not a claim

of full-source representativeness. Split buckets use the first eight hexadecimal digits of a non-price fingerprint hash modulo 100: 0–54 train, 55–69 validation, 70–84 calibration, 85–99 test. No original advertisement ID or verified vehicle identity is available. Identical non-price fingerprints may over-collapse different vehicles. Training and evaluation are separately hash-ranked before the 5,000/1,000 caps. The same fixed populations and batches are shared by fitting seeds.

**Field mapping and missingness.** JUCars uses its mileage_km, transmission, fuel_type and feat_body_condition fields. MUCars maps Mileage, Gearbox, Fuel and Condition to these four fields; mileage bands use numeric midpoints, or the stated bound for a one-ended band. Missingness means unavailable in the processed source record. Its actual cause is unknown: it may reflect collection, inapplicability or earlier concealment. Unrecoverability is a simulation stipulation, not an empirical diagnosis of those causes. Numeric inputs use training medians, missing indicators and scaling; mileage is log-transformed. Categorical missing tokens and training-fit one-hot encodings retain unknown-category handling. Targets, IDs, source indices and declared reveal aliases are excluded from predictor inputs. Sellers can conceal at most two available fields, each record receives

Table S3: Unavailable fields in the actual experimental cohorts (percent of records). Full retained and pre-cap split counts are also released in field-missingness.csv.

| Source | Cohort | $n$ | Mileage | Gearbox | Fuel | Condition |
|---|---|---|---|---|---|---|
| Synthetic | training | 5000 | 3.880 | 4.280 | 4.120 | 4.160 |
| Synthetic | evaluation | 1000 | 3.700 | 3.800 | 3.800 | 3.500 |
| JUCars | training | 4774 | 20.381 | 0.021 | 0.021 | 0.063 |
| JUCars | evaluation | 1000 | 19.900 | 0.000 | 0.000 | 0.100 |
| MUCars | training | 5000 | 0.000 | 0.000 | 0.000 | 3.500 |
| MUCars | evaluation | 1000 | 0.000 | 0.000 | 0.000 | 2.800 |

Table S4: Distribution of legal report counts (percent of records). With $m$ available fields, the count is $1 + m + \binom{m}{2}$; legal concealed-field recovery edges number $m^2$.

| Source | Cohort | 1 | 2 | 4 | 7 | 11 |
|---|---|---|---|---|---|---|
| Synthetic | training | 0.00 | 0.04 | 0.96 | 14.40 | 84.60 |
| Synthetic | evaluation | 0.00 | 0.20 | 0.70 | 12.80 | 86.30 |
| JUCars | training | 0.00 | 0.00 | 0.04 | 20.40 | 79.56 |
| JUCars | evaluation | 0.00 | 0.00 | 0.00 | 20.00 | 80.00 |
| MUCars | training | 0.00 | 0.00 | 0.00 | 3.50 | 96.50 |
| MUCars | evaluation | 0.00 | 0.00 | 0.00 | 2.80 | 97.20 |

at most one unit-cost query, failed queries spend capacity, and the primary concealment cost is zero. Extra absence or positive-cost tests of earlier models are not relabeled as RPL robustness tests.

**Synthetic generator.** Draw make $M \sim U\{0,\dots,4\}$, variant $V \sim U\{0,\dots,3\}$ and age $A \sim U\{1,\dots,18\}$. Mileage is $K = \max(100, A\exp Z)$ with $Z \sim N(\log 15000, .4^2)$. Condition $C \in \{0,1,2,3\}$ has probabilities $(.12,.28,.42,.18)$. Automatic transmission $T \sim$ Bernoulli$(.45+.06\mathbf{1}[M>2])$, diesel $D \sim$ Bernoulli$(.4)$ and fiscal power $P = 4+M+2V+U\{0,1,2\}$.

Log target is

$$y = 10.3 + .23M + .1V - .072A - .20\log(1 + K/50000) \\ + .12C + .08T + .04D + .02P - .061[C = 0, A > 10] + \epsilon, \quad \epsilon \sim N(0, .1^2).$$

Year is $2024 - A$, model identifies make/variant, and location is uniform over three labels. Four action fields are independently made unavailable with probability .04 after target generation. Shared latent attributes create feature correlations. Seeds 73101/73102 generate 6,000/1,500 train/evaluation records. This is a stipulated generator, not inferred seller behavior.

**Training schedules by experiment family.** The inherited backbone has 250 boosting iterations and no fitted neural residual when branches start. The earlier manuscript's six-initial-epoch and two-three-epoch description was incorrect; the bound production config and training receipts specify twelve residual epochs as shown below. No model or result was changed to match the corrected description.

Table S5: Residual update schedules. Optimizer state resets at each update call; epochs and report-search sweeps are different quantities.

| Experiment family | Residual calls | Training response |
|---|---|---|
| Original feedback factorial | Two calls of six epochs | One start/two sweeps, fixed epoch scores |
| Original static no-feedback | One call of twelve epochs | No query |
| Matched RPL, gap controls and loss ablation | Two calls of six epochs | No query; common report maximization |
| Search-strength intervention | Two calls of six epochs | One/two versus four/twenty; epoch/batch refresh |
| Parameter/stability variants | Explicit per-variant settings | Declared variant; no transfer of scope |
| RPL by five acquisition families | No new price-model fitting | Reused predictors; new evaluation responses |

## S2 Ten-seed recovery geometry, quotes and budget interaction

**All-edge diagnostics.** We infer all eleven report-state quotes from each of sixty existing deployment models (three sources, ten fitting seeds, $\lambda = 0, 1$ at $\beta = .5$). For every legal concealed-field recovery edge, compute $[g(T(s,a)) - g(s)]_+$ with $g(s) = [q(s) - q(H)]_+$. Reported rates use threshold $10^{-8}$ and all legal edges as denominator. Quantiles below first include zero increments within each model and are then averaged across seeds. They are not pooled-population quantiles or confidence limits. Full per-model values, including maxima, are in edge-60-models.csv. Every edge ending at $H$ has zero penalty because $g(H) = 0$ and $g(s) \geq 0$. Thus the regularizer targets intermediate recovery states. An improved positive-gain graph need not lower the raw quote on every recovery edge and does not establish truthfulness or a fixed honest anchor.

**Quote and query outcomes.** Predicted log quotes are compared with the same log target records; currency MAE is a separate metric. The MUCars honest/attacked biases move upward despite decreasing MAE. The conditional intervals on their paired shifts include zero; this is not evidence of deliberate anchor manipulation. For JUCars $\lambda = 1$, queries cover 10% of all listings,

Table S6: Ten-seed recovery diagnostics: positive-gain violations, raw quote increases and positive increment magnitudes (log units). Median and 90th percentile are zero in these source-level means except synthetic $\lambda = 0$ has mean 90th percentile .004806.

| Source | $\lambda$ | Gain viol. (%) | Quote up (%) | Mean | 95th | 99th |
|---|---|---|---|---|---|---|
| Synthetic | 0 | 15.79 | 78.38 | 0.001648 | 0.010231 | 0.030615 |
| Synthetic | 1 | 5.03 | 79.01 | 0.000663 | 0.000345 | 0.021018 |
| JUCars | 0 | 7.08 | 86.71 | 0.001498 | 0.006725 | 0.040654 |
| JUCars | 1 | 4.76 | 87.49 | 0.001032 | 0.000248 | 0.032967 |
| MUCars | 0 | 5.27 | 87.52 | 0.000743 | 0.001096 | 0.023219 |
| MUCars | 1 | 2.60 | 83.86 | 0.000393 | 0.000000 | 0.013831 |

Table S7: Ten-seed mean log quotes and signed log biases at 10% capacity. Queries are per all listings; failures in the last column are per attempted query.

| Source | $\lambda$ | $q^H$ | $q^A$ | Honest bias | Attack bias | Query (%) | Fail (%) |
|---|---|---|---|---|---|---|---|
| Synthetic | 0 | 10.420905 | 10.433338 | -0.000205 | 0.012228 | 10.00 | 25.1 |
| Synthetic | 1 | 10.416615 | 10.422780 | -0.004495 | 0.001670 | 10.00 | 36.1 |
| JUCars | 0 | 9.044371 | 9.056807 | 0.012355 | 0.024792 | 10.00 | 57.3 |
| JUCars | 1 | 9.043418 | 9.052549 | 0.011403 | 0.020534 | 10.00 | 61.7 |
| MUCars | 0 | 11.534991 | 11.541597 | 0.029095 | 0.035701 | 9.03 | 4.3 |
| MUCars | 1 | 11.540809 | 11.545842 | 0.034913 | 0.039946 | 10.00 | 3.1 |

failures 6.17% and successes 3.83%; conditional failure is 61.7%. A small change in the per-listing failure rate need not imply a small operational failure fraction. Positive log inflation is relative to the model's own honest quote; its percentage reduction is neither a percentage price cut nor realized overpayment avoided. No transaction-price outcome is observed.

**Paired budget interaction.** Let $D_b = I_{0,b} - I_{1,b}$ at fixed gap .5. We report $D_{.1} - D_0$ without interpreting it as a causal mediation effect: sellers can change their reports between budgets. Intervals independently resample ten common fitting seeds and ten common batches 5,000 times, preserving all weight/budget pairings. These post-hoc marginal percentile intervals are conditional on fixed cohorts and do not cover development-selection uncertainty or guarantee simultaneous coverage. They are distinct from the original studentized simultaneous bands. Zero-query improvements are

Table S8: Budget interaction in positive log inflation. Real-source intervals include zero.

| Source | $D_0$ | $D_{.1}$ | Interaction | Marginal 95% interval |
|---|---|---|---|---|
| Synthetic | 0.004276 | 0.006268 | 0.001992 | [0.001255, 0.002713] |
| JUCars | 0.003249 | 0.003306 | 0.000057 | [-0.000443, 0.000532] |
| MUCars | 0.001527 | 0.001573 | 0.000046 | [-0.000129, 0.000233] |

98.3% and 97.1% of the 10% improvements on JUCars/MUCars. These are descriptive ratios, not causal contribution shares. The evidence supports recovery-graph-inspired regularization with estimator effects; it does not attribute most real-source improvement to verification. For synthetic, changing primary inflation to the fixed-peer unilateral envelope changes the paired reduction from .006267693 to .006266569, a difference of $1.125 \times 10^{-6}$. This bounds that mean sensitivity without certifying a joint equilibrium.

## S3 Fixed reports versus reselected responses

For each source/seed, retain the $\lambda = 0$, 10%-capacity report profile. Recompute both models' quotes and their own scores/allocations at this fixed profile. Separately evaluate each model's saved, endogenously selected profile. For pre-query quote $q_0$, highest positive-score query without capacity $q_L$, hard-capacity quote $q_A$ and honest quote $q_H$, report signed $E = q_0 - q_H$, $R = q_L - q_0$, $C = q_A - q_L$, $G = q_A - q_H$. All 120 source/seed/weight/profile identities satisfy $E + R + C = G$ to numerical tolerance. A negative mean $G$ can coexist with positive mean $[G]_+$. Fixed reports

Table S9: Ten-seed mean signed decomposition and positive gain, log units. The reference profile is common to the two $\lambda = 0$ report modes.

| Source | $\lambda$ | Profile | $E$ | $R$ | $C$ | $G$ | $[G]_+$ |
|---|---|---|---|---|---|---|---|
| Synthetic | 0 | Reference | 0.005702 | -0.002314 | 0.009044 | 0.012432 | 0.012432 |
| Synthetic | 1 | Fixed report | -0.001008 | 0.000694 | 0.003064 | 0.002749 | 0.006503 |
| Synthetic | 1 | Reselected | 0.000569 | 0.000258 | 0.005338 | 0.006165 | 0.006165 |
| JUCars | 0 | Reference | 0.009999 | -0.005692 | 0.008130 | 0.012437 | 0.012437 |
| JUCars | 1 | Fixed report | -0.002822 | 0.002273 | -0.000980 | -0.001530 | 0.006100 |
| JUCars | 1 | Reselected | 0.006706 | -0.003770 | 0.006195 | 0.009131 | 0.009131 |
| MUCars | 0 | Reference | 0.002579 | -0.000113 | 0.004140 | 0.006606 | 0.006606 |
| MUCars | 1 | Fixed report | -0.005296 | 0.004870 | -0.001422 | -0.001848 | 0.003343 |
| MUCars | 1 | Reselected | -0.000309 | 0.001438 | 0.003904 | 0.005033 | 0.005033 |

remove report reselection but do not hold the predictor or acquisition head fixed. These are diagnostic identities, not isolated causal interventions. In particular, the real-source $R$ term does not consistently decrease under RPL; a reduction in total positive gain must not be relabeled as a direct reduction of that component.

## S4 Matched stronger-gap comparison

The new matched comparison is still running; no scientific result is asserted in this working supplement.

## S5 Numerical portability and retained failure

An independently operated Mac completed the original six synthetic/seed-13 cells. Strict numerical reproduction failed under the original tolerances; that verdict is retained. The ten-percent paired absolute improvement is about .008024 locally and .008823 externally (53.97% versus 57.54%). Direction agreement in one seed is supportive, but response quality differs: $\lambda = 1$ maximum regret changes from .0002041 to .0219727 at 10% capacity. The table distinguishes outcome direction from numerical equivalence.

Inputs and tree backbones matched; residual parameters differed. Each machine was internally repeatable in the recorded repeats and thread comparisons. The first recorded difference was an output-layer gradient; hidden forward activations were not captured, so no underlying operator is causally isolated. An isolated float64 candidate is not adopted or validated for scientific performance. We neither relax tolerances nor retroactively substitute a new scientific-equivalence criterion. No new external real-source comparison is claimed. A stronger transportability claim

Table S10: Existing external-machine numerical comparison, synthetic seed 13. Each entry is local reference / external value. Strict numerical reproduction failed under the original tolerances.

| $\lambda$ | Capacity | Inflation (log) | Honest MAE | Attacked MAE |
|---|---|---|---|---|
| 0 | 0% | 0.0175119/0.0178126 | 3585.594/3582.048 | 3720.837/3714.437 |
| 0 | 10% | 0.0148686/0.0153343 | 3585.594/3582.048 | 3688.439/3683.640 |
| 0 | 30% | 0.0128125/0.0131494 | 3585.594/3582.048 | 3665.819/3661.133 |
| 1 | 0% | 0.0121968/0.0119582 | 3615.223/3615.545 | 3676.134/3674.371 |
| 1 | 10% | 0.0068443/0.0065113 | 3615.223/3615.545 | 3634.360/3621.562 |
| 1 | 30% | 0.0041750/0.0041850 | 3615.223/3615.545 | 3616.811/3620.643 |
| $\lambda$ | Capacity | Failed query (%) | Maximum regret (log) | Nonconverged batches /10 |
| 0 | 0% | 0.00/0.00 | 0.0000000/0.0000000 | 0/0 |
| 0 | 10% | 2.70/2.50 | 0.0000000/0.0000000 | 0/0 |
| 0 | 30% | 5.40/5.40 | 0.0000000/0.0000000 | 0/0 |
| 1 | 0% | 0.00/0.00 | 0.0000000/0.0000000 | 0/0 |
| 1 | 10% | 3.80/3.90 | 0.0002041/0.0219727 | 1/1 |
| 1 | 30% | 7.60/7.40 | 0.0020809/0.0142298 | 1/2 |

requires a future locked protocol and actual independent environment, including effect and regret checks.

The public `reproduction/macos-core-v2` and `reproduction/windows-core-v1` directories contain minimal core training procedures and data routes. The archived software excludes model weights and listing-level outputs; it is not a self-contained binary model package.

## S6 Version and evidence map

The immutable software ZIP at 10.5281/zenodo.23447567 is version 1.0.0 from source `89ad56e`. It includes the earlier 90-cell crossing and external difference tables; it excludes manuscripts, large weights, raw data and listing-level outcomes. Later paper revisions and the present diagnostics are not retroactively inside that archive. Current repository citation metadata follows the author order explicitly confirmed by the authors: Tang, You (corresponding), Gu, Pang.

- Pre-review integrated paper: commit `45c2aef`; this is the earlier 28-page manuscript with the five-family crossing in its main text.
- Matched gap runner and engineering replay: commit `5fb3437`; `experiments/run_review_gap_controls_v1.py` and its immutable manifest.
- Mechanisms and cohort construction: commit `00eb579`; `notes/review-improvement-v1/`. Sixty models and 180 existing outcome cells are newly diagnosed, not newly trained.
- Current artifact hashes: `notes/review-improvement-v1/qa.json`. They bind saved files; they do not establish scientific validity.

The main text and this supplement bound claims to the observed fixed populations and declared response procedure. Fitting seeds do not sample new training populations, and independent new-market effectiveness remains unestablished.

## S7 Full removals, default scans and seed heterogeneity

The original component experiment toggles suppression scoring ($S$), feedback ($F$) and gap loss ($G$). All eight configurations retain backbone, augmentation, error and anchoring terms, with equal role separation and epoch budgets. Thus removal of two or all three components is evaluated against a trained baseline (Figure S1). These switches do not constitute three independently established innovations.

Figure S2 retains all ten fitting seeds and four nonzero path/gap comparisons, each with its same-gap zero-path reference. Negative seed effects at smaller weights remain. A separate original *three-seed* recovery-edge pilot found mean positive increments decreasing from .001790 to .000693 (synthetic), .001648 to .001048 (JUCars) and .000788 to .000450 (MUCars) at $\lambda = 1, \beta = .5$. All remained positive: zero path loss was not achieved, and the pilot is not relabeled as ten-seed evidence.

Core scans vary gap weight $0, .1, .25, .5, 1, 2$, tolerance $0, .01, .02, .05, .1$ and feedback rounds $1, 2, 4$, at 10/30% budgets. Implementation scans vary anchoring, attack weight, learning rate, epochs, head trees and width. Figures S3 and S4 explicitly mark every scanned value and engineering default; matching default observations are reused exactly, not retrained or selected on evaluation outcomes. These three-seed scans do not retune ten-seed RPL or prove an optimum. The separate availability stress adds 0/10/30% natural absence to fixed models (189 cells, 63 reused), with seven policies and fresh response search but no new fitting; it is a controlled perturbation rather than a new market or actual-seller validation.

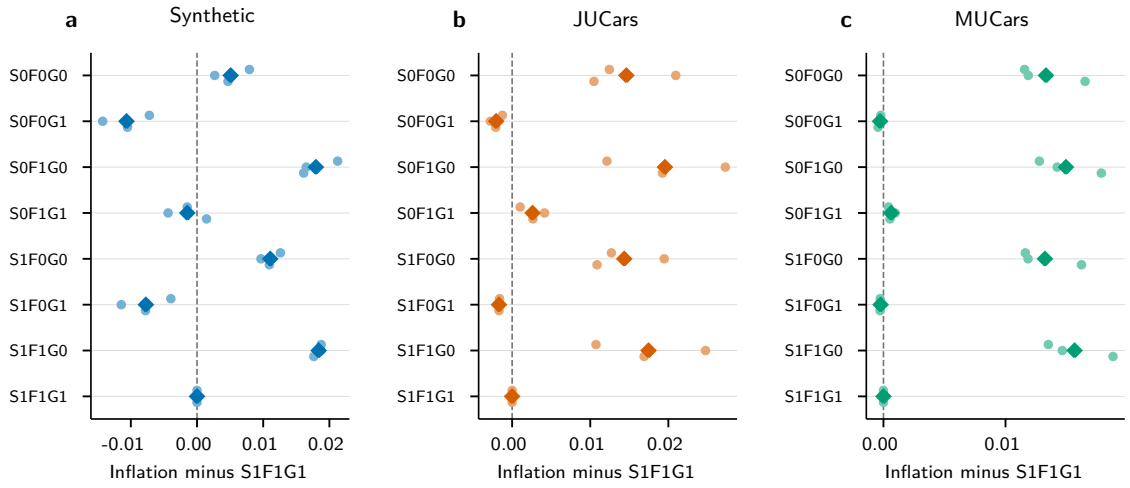
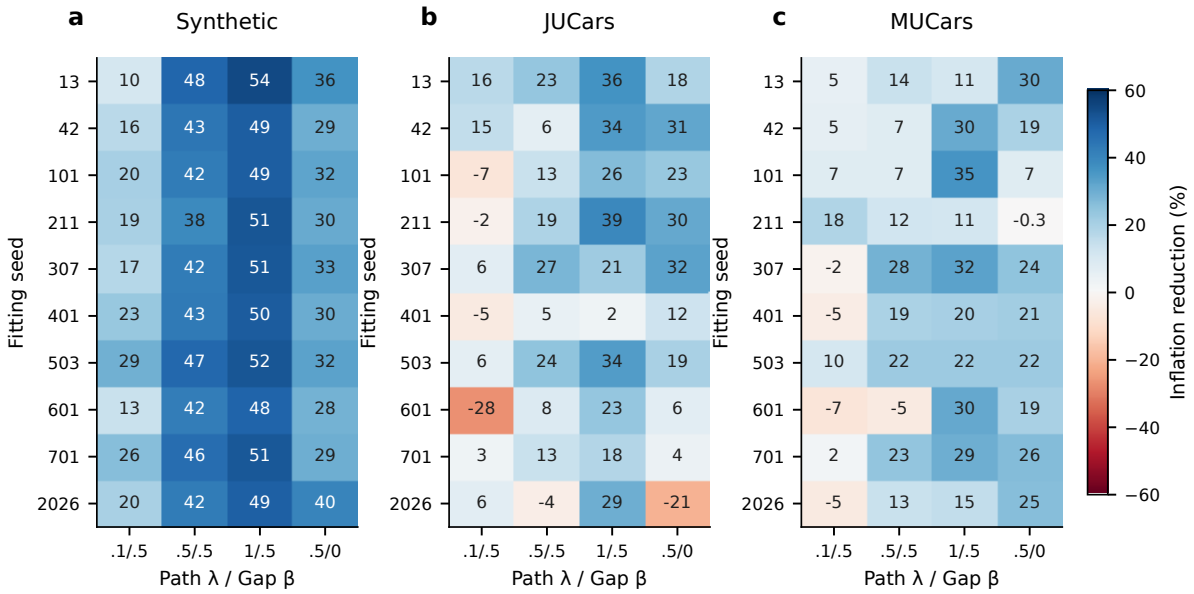



Figure S1: Complete three-component ablation at 10% capacity: circles show three paired seed differences from S1F1G1, diamonds their means. Negative values favor the ablated configuration. S/F/G denote suppression, feedback and gap switches; nonconverged responses remain.




10% capacity. Each cell is a paired seed ratio against its same-gap zero-path control.
Positive: lower inflation; negative: higher. No seed or nonconverged result is removed.

Figure S2: Ten-seed heterogeneity at 10% capacity. Positive values mean lower inflation than same-gap zero-path controls. Negative effects and nonconverged responses are retained. Labels are rounded; exact values are in the public source CSV.

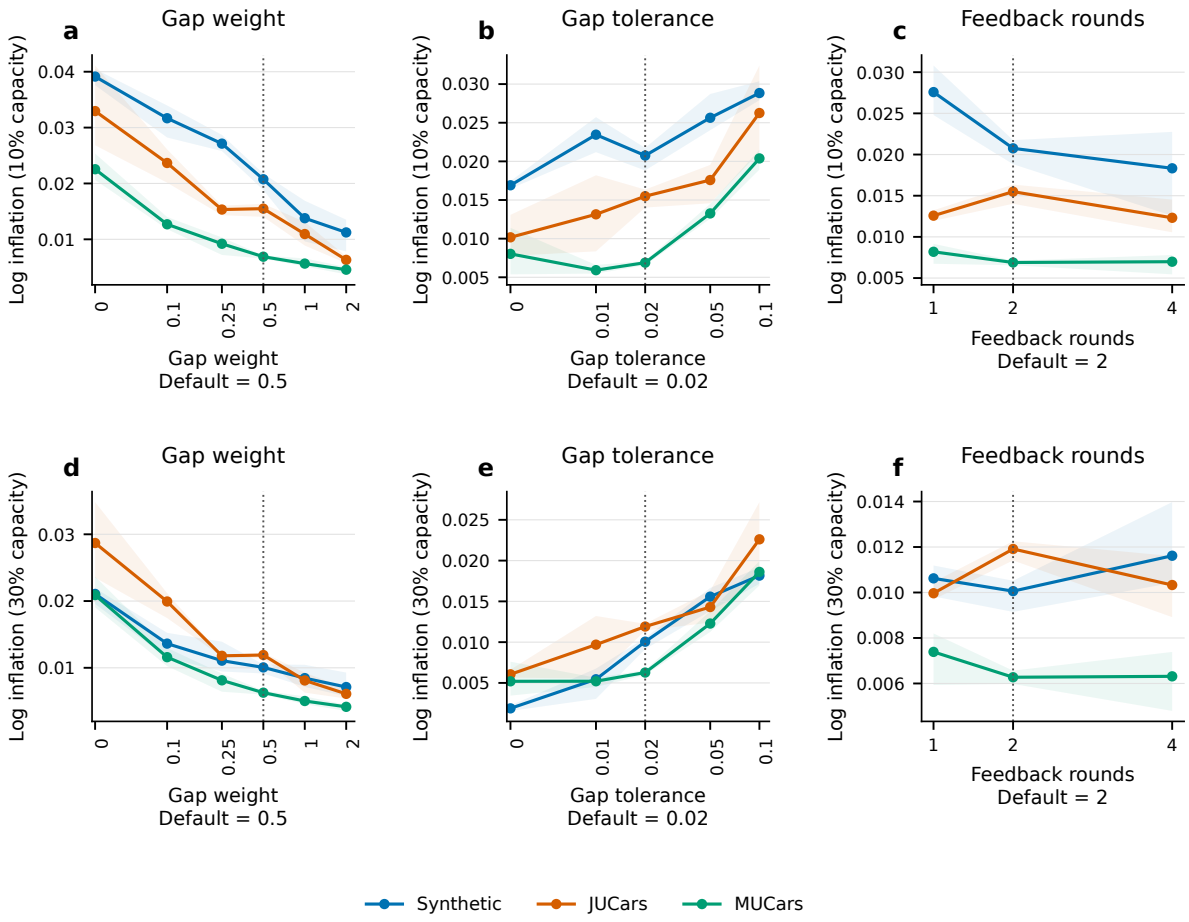



Figure S3: Core-parameter scans in the original feedback implementation, with three fitting seeds. Dotted lines mark design defaults and all evaluated settings are ticked. Gap weight and tolerance use symmetric-log coordinates. Means and full ranges are descriptive, not confidence intervals. The defaults are engineering choices, not evaluation-selected optima.

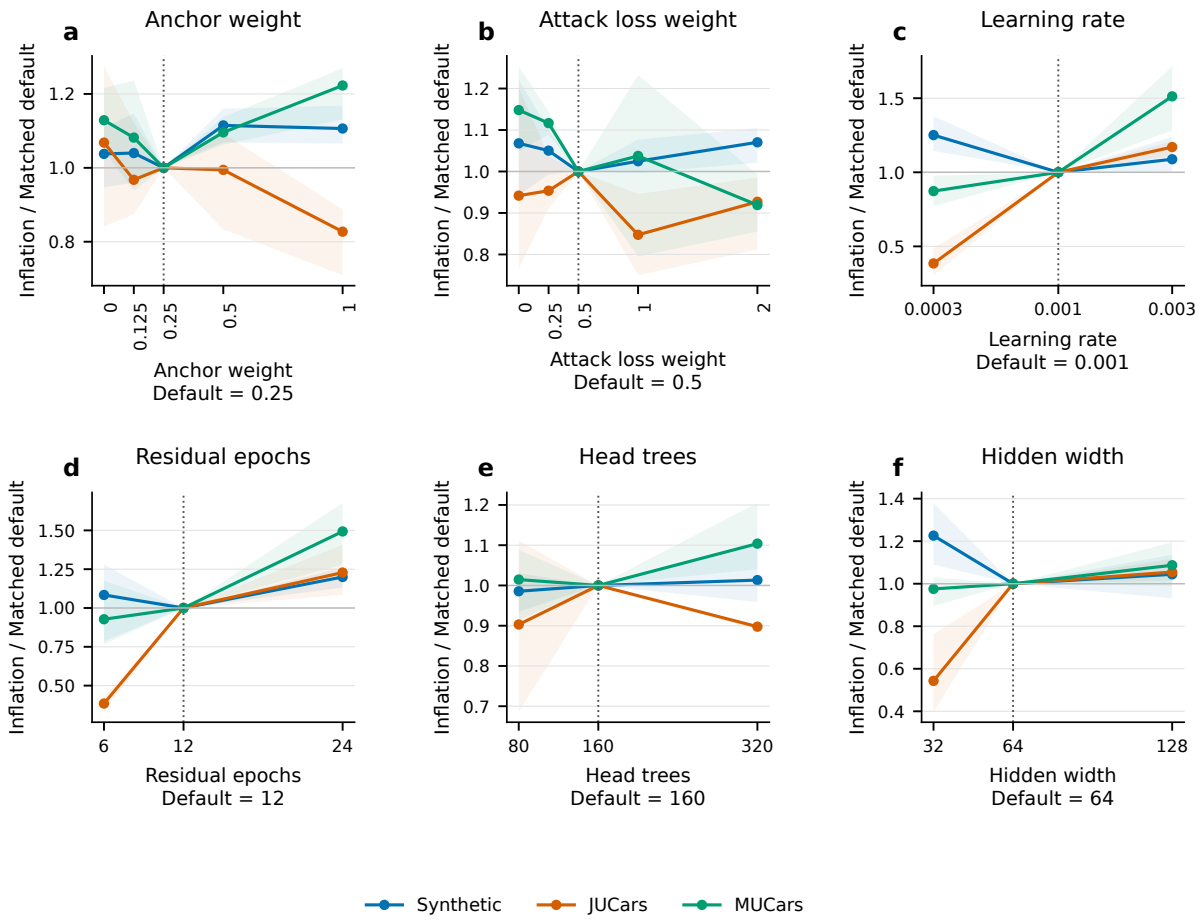



Figure S4: Six implementation-parameter scans at 10% capacity, retaining three fitting seeds. Every ratio uses the same-seed matched default. Dotted lines mark defaults; ticks show all evaluated values, and learning rate uses a logarithmic axis. Curves show means and full seed ranges, not confidence intervals.